

1. Input data

- Design shear force V_{Ed} kN (at the critical section, usually at distance d from the face of support)
- Section: b mm, d mm, h (total height) mm, $z = d - 0.5 \lambda x \approx 0.9 d$ mm
 - For T-beams also: b_{eff} mm (effective flange width), h_f mm (flange thickness)
- Longitudinal tension reinforcement area A_{sl} mm² (for calculation of ρ_l)
- Materials: f_{ck} Mpa, f_{cd} MPa, f_{yd} Mpa
- Axial design force N_{Ed} kN (positive for compression/pre-compression)
- $\nu = 0.6(1 - f_{ck}/250)$
- Gross concrete area A_c mm² = bh or $(b_{eff} - b) \cdot h_f + bh$

2. Step-by-step algorithm

Step 0 – Preliminary check: concrete shear resistance without shear reinforcement ($V_{Rd,c}$)

Compressive stress: $\sigma_{cp} = \frac{N_{Ed} 1000}{A_c}$ (Mpa) and $k_1 = 0.15$

$$C_{Rd,c} = 0.12 \text{ (Estonian NA)} \quad k = 1 + \sqrt{\frac{200}{d}} \leq 2.0 \quad \rho_l = \frac{A_{sl}}{b d} \leq 0.02$$

$$V_{Rd,c} = [C_{Rd,c} k (100 \rho_l f_{ck})^{1/3} + k_1 \sigma_{cp}] b d \geq V_{Rd,c,min} = 0.035 k^{1.5} f_{ck}^{0.5} b d$$

If $V_{Ed} \leq V_{Rd,c} \Rightarrow$ shear reinforcement not required by calculation (go to Step 3 for minimum reinforcement). If $V_{Ed} > V_{Rd,c,min} \Rightarrow$ requires check for anchorage.

If $V_{Ed} > V_{Rd,c} \Rightarrow$ shear reinforcement required (go to Step 1).

Step 1 – Determine strut inclination θ

First compute coefficient for axial compression:

$$\sigma_{cp} = 0: \alpha_{cw} = 1 \qquad 0.25 f_{cd} < \sigma_{cp} \leq 0.5 f_{cd}: \alpha_{cw} = 1.25$$

$$0 < \sigma_{cp} \leq 0.25 f_{cd}: \alpha_{cw} = 1 + \frac{\sigma_{cp}}{f_{cd}} \qquad 0.5 f_{cd} < \sigma_{cp} < 1.0 f_{cd}: \alpha_{cw} = 2.5 \left(1 - \frac{\sigma_{cp}}{f_{cd}}\right)$$

Calculate theoretical $\cot \theta$: $\cot \theta_{min} = 1$; $\cot \theta_{max} = 2.5$ (Estonian NA)

$$\cot \theta = [\cot \theta_{min} \leq \cot \left(0.5 \arcsin \left(\frac{2 V_{Ed}}{\alpha_{cw} b z \nu f_{cd}}\right)\right) \leq \cot \theta_{max}]$$

Step 2 – Calculate required shear reinforcement ratio

$$a_{sw,req} = \frac{A_{sw}}{s} = \frac{V_{Ed}}{f_{ywd} z \cot \theta}$$

Step 3 – Minimum shear reinforcement check (EC2 9.2.2(5))

$$a_{sw,min} = \frac{A_{sw,min}}{s} \geq 0.08 \frac{\sqrt{f_{ck}}}{f_{yk}} b$$

Step 4 – Final provided reinforcement

$$a_{sw,prov} = \max(a_{sw,req}; a_{sw,min})$$

Step 5 – Select stirrups and spacing

Choose stirrup diameter and number of legs, calculate A_{sw} (mm² per set).

$$s_{req} = \frac{A_{sw}}{a_{sw,prov}}$$

Select practical s such that: $s \leq s_{lim} = \min(s_{req}; 0.75 d)$

Step 6 – Verify capacity

$$V_{Rd,s} = \frac{A_{sw}}{s} z f_{yd} \cot \theta \text{ and } V_{Rd,max} = \frac{\alpha_{cw} b z v f_{cd}}{\cot \theta + \tan \theta} \text{ NB! } \alpha_{cw} \neq a_{sw}$$

Check:

$$V_{Rd,s} \geq V_{Ed} \text{ and } V_{Rd,max} \geq V_{Ed}$$